

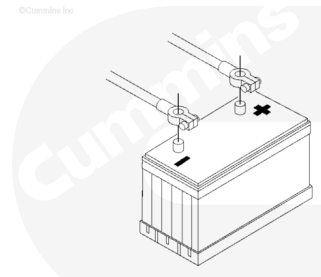
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

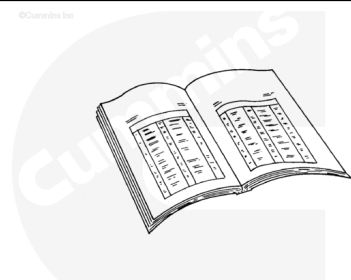
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



⚠ WARNING ⚠

Coolant is toxic. Keep away from pets and children. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

Wait until the temperature is below 50°C [120°F] to avoid personal injury from hot coolant.

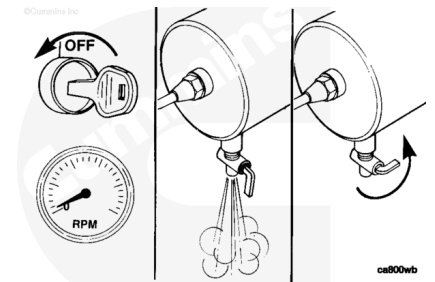
- Drain the engine coolant. Refer to Refer to Procedure Procedure 008-018
(</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>)
- Remove the coolant lines from the air compressor. Refer to Refer to Procedure Procedure 012-004
(</qs3/pubsys2/xml/en/procedures/40/40-012-004-tr.html>)
- Remove the hydraulic pump (if equipped). Refer to Refer to Procedure Procedure 009-016
(</qs3/pubsys2/xml/en/procedures/40/40-009-016.html>).

Remove

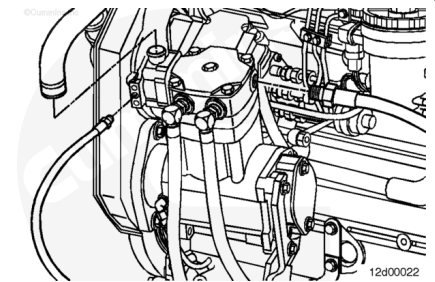
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

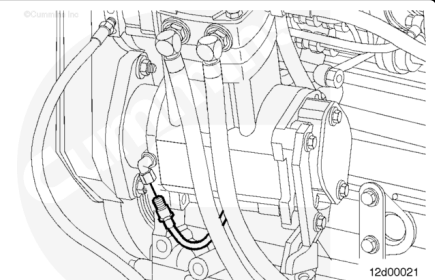
Open the draincock on the wet tank to release air from the system. Close the draincock after the pressure is released.



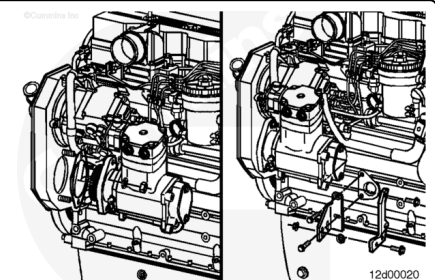
Remove the air connections from the air compressor.



Remove the oil supply line.



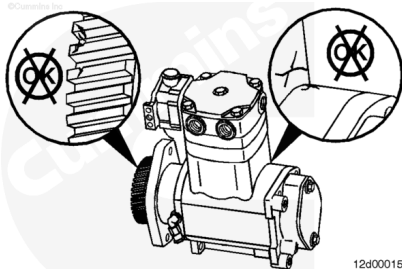
Remove the air compressor support bracket and capscrews.
Remove the air compressor mounting capscrews and the air compressor.



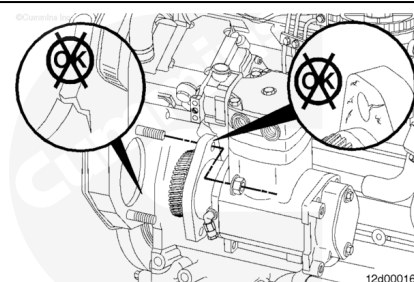
Clean and Inspect for Reuse

Inspect the compressor housing for cracks or damage.

Inspect the drive gear for cracks or damage.



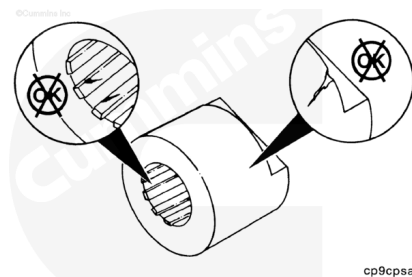
The gasket surfaces of the gear housing and air compressor **must** be clean and **not** damaged.



Power Steering Coupling (if applicable)

Inspect the coupling for wear or cracks.

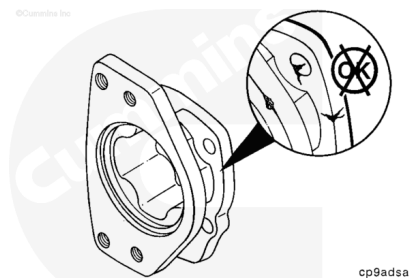
Replace the coupling, if damaged. Refer to the OEM repair manual.



Power Steering Adapter

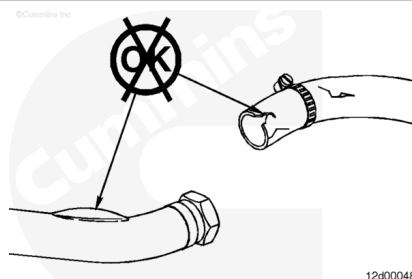
Inspect the adapter for wear or cracks.

Replace the adapter if any damage is found. Refer to the OEM repair manual.



Inspect the drive clips or connectors for damaged threads.

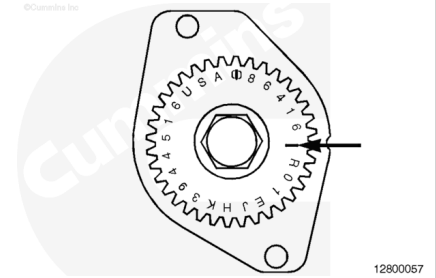
Inspect the coolant hoses and coolant lines for restrictions of coolant flow.



Install

Compressor Timing Procedure:

- Rotate the engine to Top Dead Center (TDC) using Refer to Procedure Procedure 003-004 (</qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html>).
- Rotate the gear in the air compressor until the I is in line with the V-notch on the air compressor housing, located at the 3-o'clock position as viewed from the front.



Install air compressor, two capscrews, and a new gasket onto the front gear housing.

Install the air compressor support bracket.

Torque Value:

Mounting Nuts 77 n•m [57 ft-lb]

Torque Value:

Support Capscrews to Air Compressor Support 43 n•m [32 ft-lb]

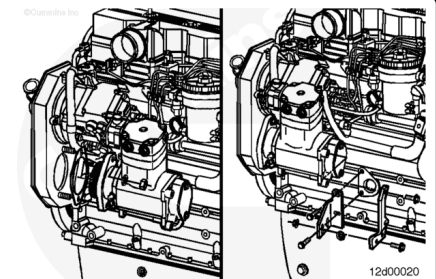
Torque Value:

Capscrews to Block 57 n•m [42 ft-lb]

Compressor drive gear mounting nut torque:

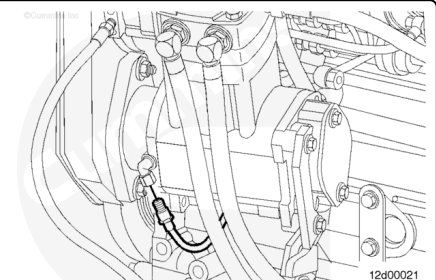
Torque Value:

Knorr-Bremse™ 165 n•m [122 ft-lb]

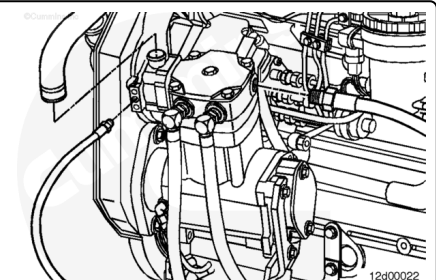


Install the oil supply line.

Torque Value: 15 n•m [133 in-lb]



Install the air lines.



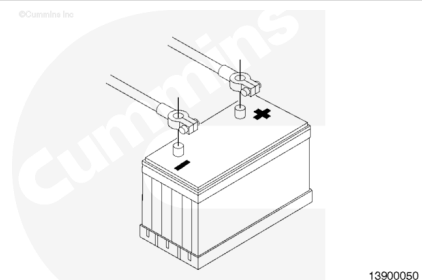
Finishing Steps

- Install the coolant lines from the air compressor. Refer to Procedure 012-004 (</qs3/pubsys2/xml/en/procedures/40/40-012-004-tr.html>)
- Install the hydraulic pump (if equipped). Refer to Procedure 009-016 (</qs3/pubsys2/xml/en/procedures/40/40-009-016.html>)
- Fill the coolant system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).



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- Connect the batteries.

- Operate the engine to check for coolant and oil leaks
- Operate the engine to actuate the air compressor
- With the air compressor pumping between 550 kPa to 690 kPa [80 psi to 100 psi], use a solution of soapy water to check for air leaks.

